Abir Modak

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Education

University of Maryland, College Park — College Park, MD
Bachelor of Science in Computer Science

- **Relevant Coursework:** Object Oriented Programming I, Object Oriented Programming II, Linear Algebra, Discrete Structures (Spring 2026), Introduction to Computer Systems (Spring 2026)

Experience

Undergraduate Researcher — Aug. 2024 – Present
Princeton University — *Princeton, NJ*

- Built instance segmentation model using Detectron2 to segment cancer tumors; achieving 91% mean IOU
- Developed a novel segmentation approach by normalizing depth information into the blue color channel; increasing IOU by 7.8%
- Created an annotation methodology with pytest validations for a series of 5000 raw images.
- Automated model training through Slurm and bash scripts, creating pipelines for 3 different approaches
- Presented key findings at the **MIT Undergraduate Research and Technology Conference**.

Frontend Engineer — Sept. 2025 – Present
TerpLabs — *College Park, MD*

- Developing a full-stack apartment subleasing platform using Next.Js; for the UMD community.
- Designed a Zillow-inspired user interface using Figma, creating a platform for students to post their subleases professionally.
- Implemented a search by location feature using Google Maps Api, serving the needs for location based apartment searching.

Frontend Engineer — Sept. 2024 – June 2025
Gradescout — *Edison, NJ*

- Contributing to a gradebook client for NJ students by webscraping using credientials, serving 70,000+ students
- Compiled edge cases across districts to optimize webscraping using Cheerio and Svelte

Projects

Data Flagging System for Mice Tumors | *Next.js, Google Sheets* — Sept. 2025 – Dec. 2025

- Created a web app with Next.js to aid in data cleaning mice tumors from training set
- Implemented multi-user functionality through state management, allowing different users to input their flags
- Integrated the Google Sheets API via a service worker for real-time, persistent data storage and easy access.

Inventory Management System | *Sveltekit, Node.js, Express.js, MongoDB* — Nov. 2023 – June 2025

- Created an inventory management system with Sveltekit, Express.js, and MongoDB to store robotics components.
- Streamlined inventory across storage rooms for tools, equipment, and parts for my First Robotics Team.
- Implemented in school's workshop, aiding students taking electrical/robotics coursework with storing components

Automated Hackathon Registration System | *Discord.js, Google Apps Script, Node.js* — Nov. 2023 – June 2025

- Automated hackathon registration with Discord.js and Google Apps Script to manage 200+ members.
- Created discord bot to query google sheets to verify users and assign roles, streamlining workflow for organizers.
- Added user interactive features, allowing participant engagement and streamlined management for organizers.

FRC Strategy Tool | *Sveltekit, Node.js* — Sept. 2022 – May 2025

- Developed a strategy web app with Sveltekit for a First Robotics team, achieving 17/60th at the state level.
- Engineered an offline-first data transfer system using QR codes, enabling reliable data collection in arena environments.
- Integrated 3rd-party data into the dashboard, enabling improved decisions alongside user collected data.

Skills

Languages: Python, C++, Java, Javasript, MATLAB
Frameworks: *AI/ML/CV:* PyTorch, Detectron2, OpenCV, YOLO; ***Web:*** Figma, Node.js, React.js, SvelteKit, Express.js
Developer Tools: Git, GitHub, Conda, Linux, Slurm (sbatch), Raspberry Pi, Arduino